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QUARTERLY EXAMINATION - 2025	12 - STD	
MATHEMATICS	Marks 90	Time 3.00 Hrs.

PART - I

I. Answer all the questions. **20 x 1 = 20**

1. If A is a non-singular matrix such that $A^{-1}=\begin{bmatrix}5 & 3\\ -2 & -1\end{bmatrix}$, then $(A^T)^{-1}$ = ...

(a) $\begin{bmatrix}-5 & 3\\ 2 & 1\end{bmatrix}$ (b) $\begin{bmatrix}5 & 3\\ -2 & -1\end{bmatrix}$ (c) $\begin{bmatrix}-1 & -3\\ 2 & 5\end{bmatrix}$ (d) $\begin{bmatrix}5 & -2\\ 3 & -1\end{bmatrix}$

2. If $|\text{adj (adj A)}| = |A|^9$, then the order of the square matrix A is .....

a) 3 b) 4 c) 2 d) 5

3. If $0 \le \theta \le \pi$ and the system of equations $x + (\sin\theta)y - (\cos\theta)z = 0$, $(\cos\theta)x - y + z = 0$, $(\sin\theta)x + y - z = 0$ has a non-trivial solution then θ is

a) $\frac{2\pi}{3}$ b) $\frac{3\pi}{4}$ c) $\frac{5\pi}{6}$ d) $\frac{\pi}{4}$

4. If $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is .....

a) $\frac{1}{2}$ b) 1 c) 2 d) 3

5. If $|z-2+i| \le 2$, then the greatest value of $|z|$ is .....

a) $\sqrt{3}-2$ b) $\sqrt{3}+2$ c) $\sqrt{5}+2$ d) $\sqrt{5}-2$

6. The product of all four values of $\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is

a) -2 b) -1 c) 1 d) 2

7. A polynomial equation in x of degree n always has .....

a) n distinct roots b) n real roots c) exactly n complex roots d) atmost one root

8. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is .....

a) 2 b) 4 c) 1 d) ∞

9. The number of positive zeros of the polynomial $\sum_{r=0}^{n} nc_r(-1)^r x^r$ is .....

a) 0 b) n c) $< n$ d) r

10. $\sin^{-1}\frac{3}{5}-\cos^{-1}\frac{12}{13}+\sec^{-1}\frac{5}{3}-\text{cosec}^{-1}\frac{13}{12}$ is equal to .....

a) 2π b) π c) 0 d) $\tan^{-1}\frac{12}{65}$

11. If $\cot^{-1}2$ and $\cot^{-1}3$ are two angles of a triangle then the third angle is .....

a) $\frac{\pi}{4}$ b) $\frac{3\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{\pi}{3}$

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12. If $sin^{-1}\frac{x}{5}+cosec^{-1}\frac{5}{4}=\frac{\pi}{2}$, then the value of x is .....

a) 4 b) 5 c) 2 d) 3

13. The radius of the circle $3x^2+by^2+4bx-6by+b^2=0$ is .....

a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

14. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is .....

a) 2ab b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

15. The locus of a point whose distance from (-2, 0) is $\frac{2}{3}$ times its distance from the line $x=\frac{-9}{2}$ is .....

a) a parabola b) a hyperbola c) an ellipse d) a circle

16. If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}\,\vec{c}\,\vec{b}]$ is .....

a) 2 b) -1 c) 1 d) 0

17. If $\vec{a},\vec{b},\vec{c}$ are non-coplanar, non zero vectors such that $[\vec{a}\,\vec{b}\,\vec{c}]=3$, then $\{[\vec{a}\times\vec{b},\vec{b}\times\vec{c},\vec{c}\times\vec{a}]\}^2$ is .....

a) 81 b) 9 c) 27 d) 18

18. Distance from the origin to the plane $3x-6y+2z+7=0$ is

a) 0 b) 1 c) 2 d) 3

19. If $y=2\sqrt{2}x+c$ is a tangent to the circle $x^2+y^2=16$, find c.

$c^2=a^2(1+m^2)$

Ans: ± 12

a) $\pm 2\sqrt{2}$ b) ± 8 c) $\pm\sqrt{3}$ d) ± 64

20. Which one of the points i, $-2+i$ and 3 is farthest from the origin?

$|z|=\sqrt{x^2+y^2}$

Example 2.11

a) 3 b) $-2+i$ c) i d) 2

PART - II

Answer any seven questions. Q.No. 30 is compulsory. **7 x 2 = 14**

21. If $adj\,A=\begin{bmatrix}-1 & 2 & 2\\ 1 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$, Find A^{-1} Example 1.6 page No:9

22. Simplify $\sum_{n=1}^{12} i^n$ Exercise 2.1 (3) page No: 54

23. Find a polynomial equation of minimum degree with rational co-efficients having $2-\sqrt{3}$ as a root. Example 3.9 page No: 110

24. Find the principal value of $sin^{-1}(2)$, if it exists. Example 4.2 page No: 136

25. If $y=4x+c$ is a tangent to the circle $x^2+y^2=9$, find c. Example 5.12 page No: 181

26. If $2\hat{i}-\hat{j}+3\hat{k}, 3\hat{i}+2\hat{j}+\hat{k}, \hat{i}+m\hat{j}+4\hat{k}$ are coplanar, find the value of m.

Example 6.15 page No: 236

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27. If α,β and γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \frac{1}{\beta\gamma}$ in terms of the coefficients. Example 3.3 page No: 104

28. Find the vertices, foci for the hyperpola $9x^2 - 16y^2 = 144$. Example 5.25 page No: 195

29. Find the angle between the straight line $\vec{r} = (2\hat{i}+3\hat{j}+\hat{k}) + t(\hat{i}-\hat{j}+\hat{k})$ and the plane $2x - y + z = 5$. Example 6.48 Page No: 268

30. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal. Example 1.11 page No: 13

PART - III

III. Answer any seven questions. Q.No. 40 is compulsory. **7 x 3 = 21**

31. Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ Example 1.17 Page No: 22

32. If $|z| = 3$, show that $7 \le |z + 6 - 8i| \le 13$ Exercise 2.5 (4) page No: 72

33. Solve the equation $x^3 - 3x^2 - 33x + 35 = 0$. Example 3.17 Page No: 115

34. Find the value of $\tan^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{-1}{2}\right)$ Example 4.10 Page No: 147

35. Obtain the equation of the circle for which (3, 4) and (2, -7) are the ends of a diameter. Exercise 5.1 (5) Page No: 182

36. Find the distance between the parallel planes $x + 2y - 2z + 1 = 0$ and $2x+4y-4z+5 = 0$. Example 6.51 Page No: 271

37. Find the exact number of real zeros and imaginary of the polynomial $x^9 + 9x^7 + 7x^5 + 5x^3 + 3x$. Exercise 3.6 (5) Page No: 127

38. Obtain the cartesian equation for the locus of $z = x + iy$ in $|z| = |z-i|$. Example 2.21 Page No: 74

39. Find : $\sqrt{4+3i}$ Exercise 2.5 10(i) Page No: 72

40. If $\vec{a} = -3\hat{i} - \hat{j} + 5\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{c} = 4\hat{j} - 5\hat{k}$ find $\vec{a}.(\vec{b}\times\vec{c})$. Example 6.12 page No: 235

PART - IV

JV. Answer all the questions. **7 x 5 = 35**

41. a) By using Gaussian elimination method, balance the chemical reaction
$C_2H_6 + O_2 \rightarrow H_2O + CO_2$ Exercise 1.7 Page No: 47

(OR)

b) Given the complex number $z = 3 + 2i$, represent the complex number z, iz, and $z + iz$ in one Argand plane. Show that these complex number form the vertices of an isosceles right triangle. Example 2.18 Page No: 73

42. a) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution. Exercise 3.5 (7) Page No: 124

(OR)

b) Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ Example 4.28 Page No: 165

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43. a) On lighting a rocket cracker it gets projected in a parabolic path and reaches maximum height of 4m when it is 6m away from the point of projection. Finally it reaches the ground 12m away from the starting point. Find the angle of projection.

(OR) Exercise 5.5(9) Page No: 215

b) If $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}+5\hat{j}+2\hat{k}$, $\vec{c}=-\hat{i}-2\hat{j}+3\hat{k}$ verify that $(\vec{a}\times\vec{b})\times\vec{c}=(\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a}$

Exercise 6.3 (4)(i) Page No: 242

44. a) Find the fourth roots of unity. Example 2.33 page No: 89

(OR)

b) Find the equation of the circle passing through the points (1, 1), (2, -1) and (3, 2).

Example 5.10 page No: 178

45. a) If $A=\begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$ verify that $A(adj\ A) = (adj\ A)\ A = |A|I_2$

(OR) Exercise 1.1 (6) page No: 16

b) Find the centre, foci and eccentricity of the hyperbola
$11x^2 - 25y^2 - 44x + 50y - 256 = 0$ Example 5.26 page No: 195

46. a) Find the non-parametric form of vector equation and cartesian equations of the plane passing through the points (2, 2, 1), (9, 3, 6) and perpendicular to the plane $2x + 6y + 6z = 9$. Exercise 6.7 (2) page No: 263

(OR)

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$.

Example 4.22 page No: 163

47. a) If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation
$x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x - 140 = 0$. Find all roots.

(OR) Example 3.15 page No: 114

b) Prove by vector method $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

Example 6.5 page No: 227

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